

```
In [1]: import warnings
warnings.filterwarnings('ignore')
```

```
In [2]: # !pip install langchain
```

```
In [3]: # pip install openai
```

```
In [4]: # !pip install pypdf
```

```
In [5]: # !pip install tiktoken
```

```
In [6]: # !pip install faiss-cpu
```

```
In [7]: # !pip install sentence-transformers
```

```
In [8]: # !pip install tf-keras
```

```
In [9]: import os
from dotenv import load_dotenv
load_dotenv()
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

```
In [10]: from langchain_community.document_loaders import PyPDFLoader

loader = PyPDFLoader("data/The_Adventures_of_Tom_Sawyer.pdf")
document = loader.load()
document[5].page_content[:5000]
```

WARNING:tensorflow:From C:\ProgramData\anaconda3\Lib\site-packages\tf_keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sparse_softmax_cross_entropy instead.

```
Out[10]: 'Chapter 1    The Fence \n \nTom Sawyer lived with his aunt because his mother and \nfather were dead. Tom didn't like going to school, and he \ndidn't like working. He liked playing and having \nadventures. One Friday, he didn't go to school—he went \nto the river. \nAunt Polly was angry. “You’re a bad boy!” she said. \n“Tomorrow you can’t play with your friends because yo u \ndidn’t go to school today. Tomorrow you’re going to work \nfor me. You can paint the fence.” \nSaturday morning, Tom was not happy, but he started to \npaint the fence. His friend Jim was in the street. \nTom asked him, “Do you want to paint?” \nJim said, “No, I can’t. I’m going to get water.” \nThen Ben came to Tom’s house. He watched Tom and \nsaid, “I’m going to swi m today. You can’t swim because \nyou’re working.” \nTom said, “This isn’t work. I like painting.” \n“Can I paint, too?” Ben asked. \n“No, you can’t,” Tom answered. “Aunt Polly asked me \nbecause I’m a very good painter.” \nBen said, “I’m a good pain ter, too. Please, can I paint? I \nhave some fruit. Do you want it?” \nOK,” Tom said. “Give me the fruit. Then you can pain t.” \nBen started to paint the fence. Later, many boys came to \nTom’s house. They watched Ben, and they wanted to \npaint, t oo. \nTom said, “Give me some food and you can paint.” \n \n1'
```

```
In [11]: from langchain_classic.vectorstores import FAISS
from langchain_classic.embeddings import OpenAIEmbeddings
```

```
embeddings = OpenAIEmbeddings() #임베딩 처리
db = FAISS.from_documents(document, embeddings)
```

C:\Users\User\AppData\Local\Temp\ipykernel_22020\516027925.py:4: LangChainDeprecationWarning: The class `OpenAIEmbeddings` was deprecated in LangChain 0.0.9 and will be removed in 1.0. An updated version of the class exists in the `langchain-openai` package and should be used instead. To use it run `pip install -U `langchain-openai` and import as `from `langchain_openai import OpenAIEmbeddings``.

```
embeddings = OpenAIEmbeddings() #임베딩 처리
```

```
In [12]: text = "진희는 강아지를 키우고 있습니다. 진희가 키우고 있는 동물은?"
text_embedding = embeddings.embed_query(text)
print(text_embedding[:50])
```

```
[-0.0024528843509084284, -0.020228803288418053, -0.012871727430483064, -0.016594432401514297, -0.022462425898507762, 0.02859542
537317398, -0.027888741524283195, 0.0066756308509792455, -0.01205777943469094, 0.00996927923440513, -0.01558488524337078, 0.014
171519408234372, 0.00695956642576457, -0.03326458132883371, 0.0006356992567025178, 0.028090652259763513, 0.02251290358237784, -
0.005038271690653234, 0.02386317231267665, -0.02715682069610253, -0.00023148601907757777, 0.003754253756280678, 0.0142598544236
8443, -0.021049060063371355, -0.004893149280849337, 0.01820970897213102, 0.024367945426087115, -0.020670479296990923, -0.004016
105111542197, -0.01863876555973637, 0.001209090588039197, -0.0013147775401502397, -0.035359390308280704, -0.00519916837666653,
0.0011941050256402675, -0.01781850878478307, 0.010915729287711048, 0.007022663064940878, 0.009458195479188322, 0.00346085408143
1005, -0.000671191145343022, 0.0033251962373527484, 0.005429471412339895, -0.007565294441253905, 0.015685839679788358, 0.004880
529859881817, -0.012240760107397398, -0.021112157168208953, -0.019358068829594673, 0.014752009047449959]
```

```
In [14]: from langchain_classic.embeddings import HuggingFaceEmbeddings
embeddings = HuggingFaceEmbeddings(model_name = "sentence-transformers/all-MiniLM-L6-v2")
```

```
text = "진희는 강아지를 키우고 있습니다. 진희가 키우고 있는 동물은?"
text_embedding = embeddings.embed_query(text)
print(text_embedding[:50])
```

C:\Users\User\AppData\Local\Temp\ipykernel_22020\2091208629.py:2: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated in LangChain 0.2.2 and will be removed in 1.0. An updated version of the class exists in the `langchain-huggingface` package and should be used instead. To use it run `pip install -U `langchain-huggingface` and import as `from `langchain_huggingface import HuggingFaceEmbeddings``.

```
embeddings = HuggingFaceEmbeddings(model_name = "sentence-transformers/all-MiniLM-L6-v2")
[0.013278781436383724, 0.07225914299488068, 0.09263104945421219, -0.003979558125138283, 0.0015618313336744905, -0.1030637174844
7418, 0.10929880291223526, 0.05566200613975525, -0.03116736002266407, -0.05020312964916229, 0.08312954008579254, -0.00892439484
5962524, 0.09506326168775558, -0.06980784237384796, 0.03955903649330139, -0.10899197310209274, 0.04943859949707985, 0.037364806
97989464, -0.12409231066703796, -0.003315398469567299, 0.04840955510735512, -0.031085066497325897, 0.008206980302929878, 0.0632
6042860746384, -0.06804246455430984, -0.010208171792328358, 0.0049270582385361195, -0.014940373599529266, -0.001476580277085304
3, -0.0065989443100988865, -0.04015948623418808, 0.08289807289838791, 0.014144627377390862, -0.011793523095548153, -0.094151377
67791748, 0.002156355883926153, -0.019053084775805473, -0.03773903846740723, -0.003271045396104455, 0.04685607925057411, -0.181
11620843410492, -0.11718786507844925, 0.03504841774702072, -0.06848111748695374, 0.06553443521261215, 0.03522855415940285, -0.0
49748484045267105, -0.126129612326622, -0.0784597247838974, 0.004257856402546167]
```

```
In [15]: from langchain_classic.chat_models import ChatOpenAI
llm = ChatOpenAI(temperature=0, # 창의성 0으로 설정
                model_name='gpt-4.1-mini', # 모델명
                )

from langchain_classic.chains import RetrievalQA
retriever = db.as_retriever()

qa = RetrievalQA.from_chain_type(
    llm=llm,
    chain_type="stuff",
    retriever=retriever)

query = "마을 무덤에 있던 남자를 죽인 사람은 누구니?"
result = qa({"query": query})
print(result['result'])
```

```
C:\Users\User\AppData\Local\Temp\ipykernel_22020\4182033347.py:2: LangChainDeprecationWarning: The class `ChatOpenAI` was deprecated in LangChain 0.0.10 and will be removed in 1.0. An updated version of the class exists in the `langchain-openai` package and should be used instead. To use it run `pip install -U `langchain-openai` and import as `from `langchain_openai import ChatOpenAI``.
```

```
    llm = ChatOpenAI(temperature=0, # 창의성 0으로 설정
```

```
C:\Users\User\AppData\Local\Temp\ipykernel_22020\4182033347.py:15: LangChainDeprecationWarning: The method `Chain.__call__` was deprecated in langchain-classic 0.1.0 and will be removed in 1.0. Use `invoke` instead.
```

```
    result = qa({"query": query})
```

마을 무덤에 있던 남자를 죽인 사람은 인준 조(Injun Joe)입니다.

In []: